



IST687 FINAL PROJECT

WINE QUALITY

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IST687 Intro to Data Science
Syracuse University
Winter 2024



Syracuse
University

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Introduction



Portuguese “Vinho Verde”

Wine quality is intricate due to the interaction of intrinsic and extrinsic factors in evaluations. Subjective perceptions add to the challenge of building an accurate rating system. Quality is assessed through sensory or physicochemical tests, with the former relying on human senses for evaluation. **A 2009 study used data mining to predict human wine preferences based on analytical data, aiming to improve wine makers' decision-making efficiency.** A similar study by Verissimo analyzed the correlation between red wine's chemical composition and sensory perception, focusing on visual attributes and sensory characteristics. While other studies have explored specific taste aspects and perception, **we chose to assess if the chemical composition of wines correlates with their success with sommeliers, which has practical implications for winemakers and companies.** Quality scores from 1 to 10 were used as the independent variable, representing the median value of three gradings given by independent judges for each wine sample.. The dataset comprises 6497 observations, each with 13 variables. **Our primary focus was on assessing wine quality within this dataset. It was interesting to uncover which features cause the most influence on this quality metric.**

BUSINESS QUESTIONS

WHITE



RED



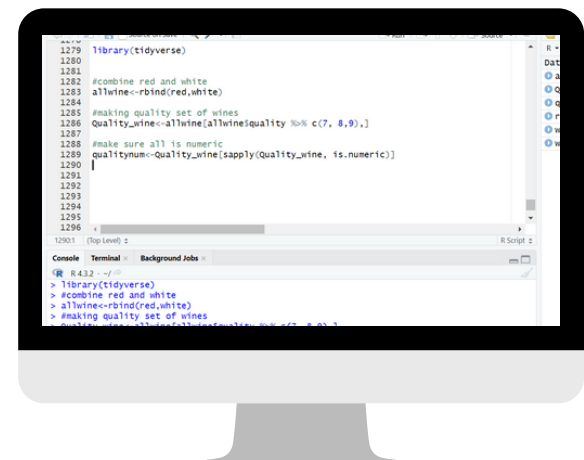
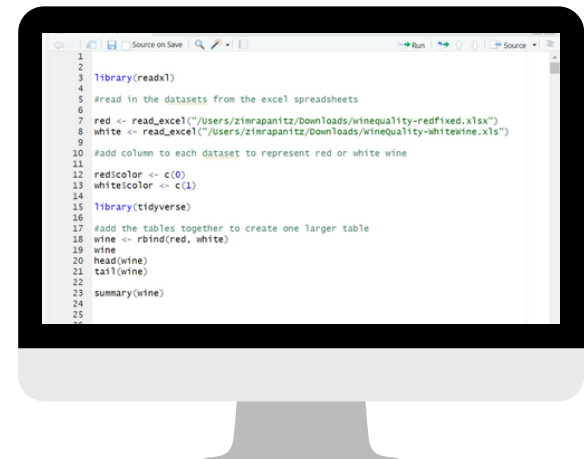
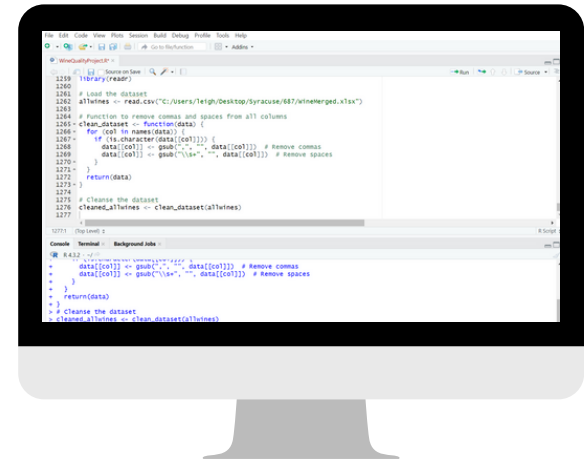
1. Differentiate the characteristics of high-quality red and white wines (those with quality 7-10). What distinguishes a red wine as high-quality and a white wine as high-quality?
2. What are the primary factors that influence the taste quality (those with quality 7-10) of red and white wines?
 - a. After identifying these factors, specify which one the company should prioritize to enhance or maintain the high-quality taste of each wine.

DATA CLEANSE

CLEANSE

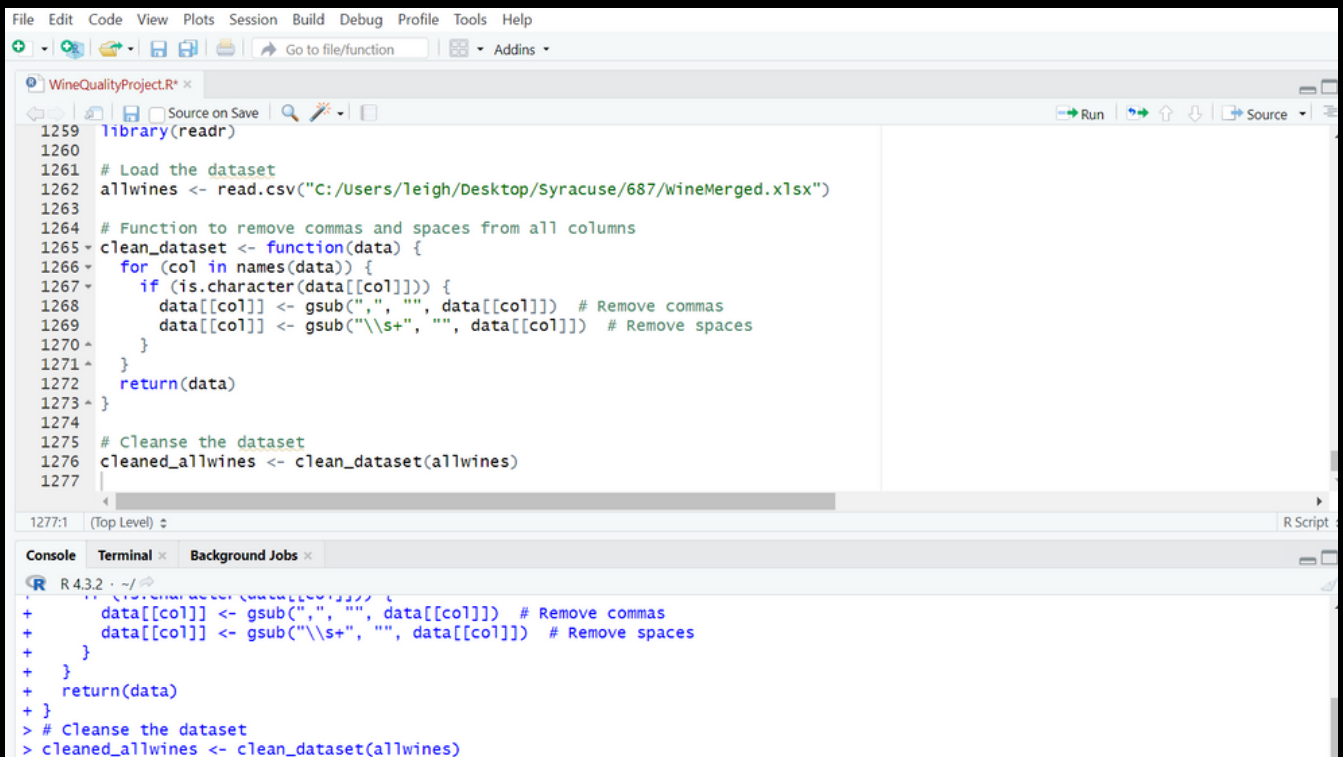
MUNGE

PREPARE



CLEANSE

Cleansing the data involves identifying and correcting any inconsistencies, errors, or missing values within the dataset. This includes tasks such as removing duplicate records, fixing any typographical errors, filling in any missing values, and ensuring that all data types are in a consistent and suitable format for analysis.



The screenshot displays the RStudio interface with a script editor and a console. The script editor shows the following R code:

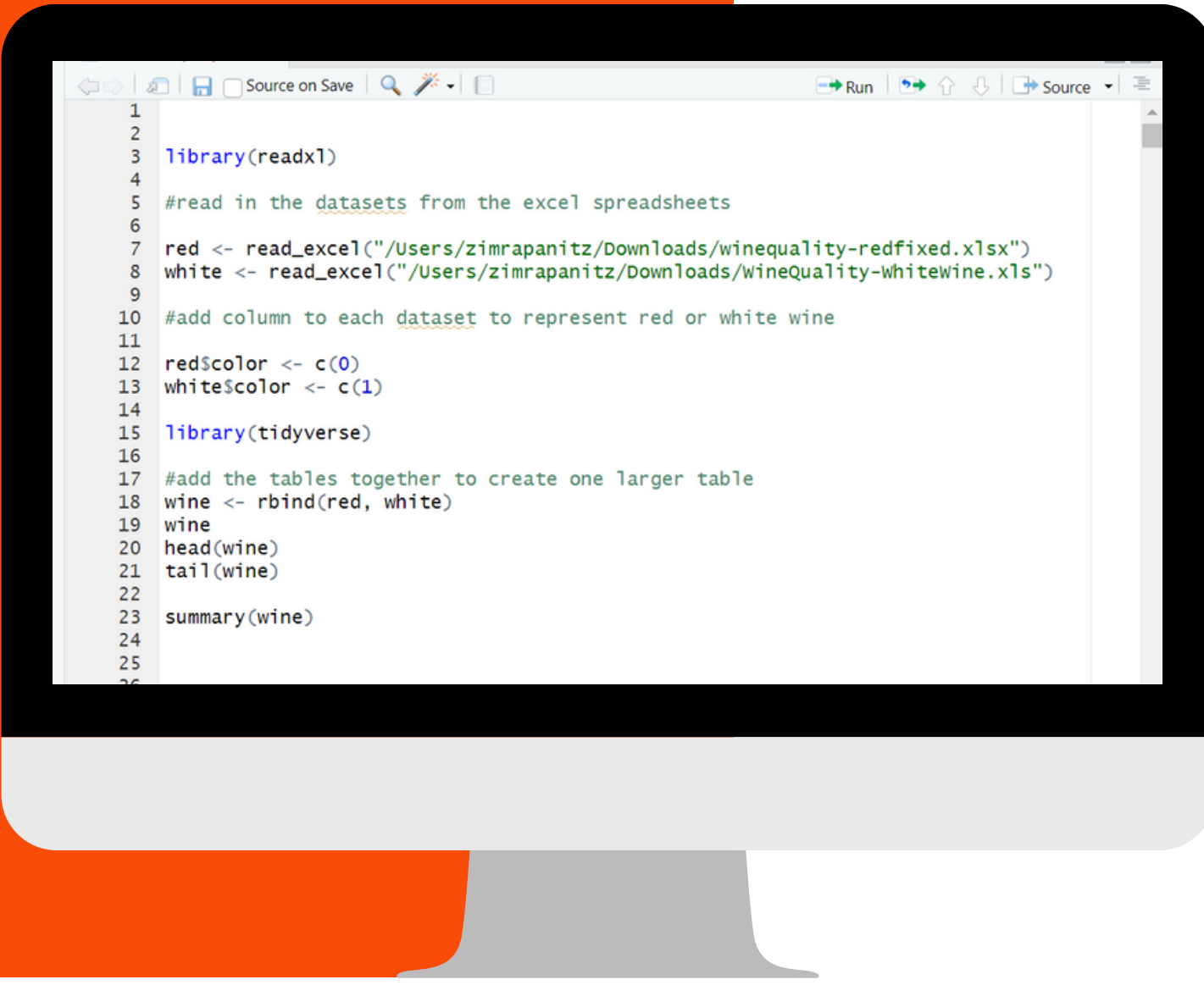
```
1259 library(readr)
1260
1261 # Load the dataset
1262 allwines <- read.csv("C:/Users/leigh/Desktop/Syracuse/687/WineMerged.xlsx")
1263
1264 # Function to remove commas and spaces from all columns
1265 clean_dataset <- function(data) {
1266   for (col in names(data)) {
1267     if (is.character(data[[col]])) {
1268       data[[col]] <- gsub(",", "", data[[col]]) # Remove commas
1269       data[[col]] <- gsub("\\s+", "", data[[col]]) # Remove spaces
1270     }
1271   }
1272   return(data)
1273 }
1274
1275 # Cleanse the dataset
1276 cleaned_allwines <- clean_dataset(allwines)
1277
```

The console shows the execution of the code, with the following output:

```
R 4.3.2 ~ /
+   if (is.character(data[[col]])) {
+     data[[col]] <- gsub(",", "", data[[col]]) # Remove commas
+     data[[col]] <- gsub("\\s+", "", data[[col]]) # Remove spaces
+   }
+ }
+ return(data)
+ }
> # Cleanse the dataset
> cleaned_allwines <- clean_dataset(allwines)
```

MUNGE

A column for red and white wine was added to each database, with "0" and "1" respectively. Subsequently, the two datasets were concatenated using `rbind`. This procedure was successful, resulting in red, white, and wine datasets, each comprising 13 variables and being presented as tibbles.



```
1  
2  
3 library(readxl)  
4  
5 #read in the datasets from the excel spreadsheets  
6  
7 red <- read_excel("/Users/zimrapanitz/Downloads/winequality-redfixed.xlsx")  
8 white <- read_excel("/Users/zimrapanitz/Downloads/wineQuality-whitewine.xls")  
9  
10 #add column to each dataset to represent red or white wine  
11  
12 red$color <- c(0)  
13 white$color <- c(1)  
14  
15 library(tidyverse)  
16  
17 #add the tables together to create one larger table  
18 wine <- rbind(red, white)  
19 wine  
20 head(wine)  
21 tail(wine)  
22  
23 summary(wine)  
24  
25  
26
```

PREPARE

The final step involved sorting the dataset based on wine quality, and wines with a quality rating of 7-10 were split into their own data set.

```
1279 library(tidyverse)
1280
1281
1282 #combine red and white
1283 allwine<-rbind(red,white)
1284
1285 #making quality set of wines
1286 Quality_wine<-allwine[allwine$quality %>% c(7, 8,9),]
1287
1288 #make sure all is numeric
1289 qualitynum<-Quality_wine[sapply(Quality_wine, is.numeric)]
1290 |
1291
1292
1293
1294
1295
1296
```

1290:1 (Top Level) ↕

R Script ↕

Console Terminal × Background Jobs ×

R 4.3.2 · ~/

```
> library(tidyverse)
> #combine red and white
> allwine<-rbind(red,white)
> #making quality set of wines
> Quality_wine<-allwine[allwine$quality %>% c(7, 8,9),]
```


ATTRIBUTES ANALYZED

FIXED ACIDITY

Fixed acidity in wine refers to the non-volatile acids present, such as tartaric, malic, citric, and succinic acids, which contribute to the wine's tartness and freshness. It is measured in grams per liter (g/L) or milliequivalents per liter (meq/L).

VOLATILE ACIDITY

Volatile acidity (VA) in wine refers to the presence of volatile organic acids, primarily acetic acid, that can be detected by their aroma. High levels can indicate spoilage or fermentation issues, and VA can be measured by titration.

CITRIC ACID

Citric acid in wine is a natural organic acid found in grapes, contributing to freshness and tartness. It can also be added during winemaking to increase acidity or balance flavors. Levels vary based on grape variety and growing conditions.

RESIDUAL SUGAR

Residual sugar (RS) in wine refers to the natural sugars that remain after fermentation. It can come from grapes or be added, and its level can vary from dry to sweet. Residual sugar affects taste, mouthfeel, and sweetness, measured in g/L or as a percentage of the wine's total volume.

CHLORIDES

Chlorides in wine are chloride ions (Cl-) that can come from grapes, water, or additives. Their levels vary based on factors like grape variety and winemaking practices, impacting taste. High levels can suggest poor winemaking or contamination.

ALCOHOL

Alcohol in wine is the result of fermenting grape sugars into ethanol. It influences body, texture, and mouthfeel, and is measured as a percentage by volume (ABV). Alcohol acts as a preservative, safeguarding wine from spoilage and oxidation.

FREE SULPHUR DIOXIDE

Fixed sulfur dioxide (SO₂) in wine is chemically bound to other wine components, primarily sugars and phenols, and is essential for preventing oxidation and microbial spoilage. It cannot be measured by standard methods, unlike free SO₂.

BOUND SULPHUR DIOXIDE

Bound sulfur dioxide (SO₂) in wine refers to the portion of SO₂ that is chemically bound to other wine components, primarily sugars and phenols. It contributes to the wine's antioxidant and antimicrobial properties but cannot be measured by standard methods.

DENSITY

Density in wine is the mass per unit volume, usually measured in grams per milliliter or kilograms per liter. It can be influenced by sugar and alcohol content, with higher density indicating a fuller body.

PH

pH in wine measures its acidity or alkalinity on a scale of 0 to 14, with 7 being neutral. A pH below 7 indicates acidity, while above 7 indicates alkalinity. pH affects the taste, balance, and stability of wine, with typical ranges between 3 and 4.

SULPHATES

Sulphates, or sulfur dioxide (SO₂), are compounds added to wine during production to prevent oxidation and microbial spoilage, preserving freshness and stability. However, excessive sulphate levels can cause adverse health effects, so winemakers must carefully regulate and monitor their use.

QUALITY

Quality in wine refers to its overall excellence, including factors like taste, aroma, balance, complexity, and structure. It is determined by grape variety, terroir, winemaking process, and aging potential, and can be assessed through sensory analysis, expert ratings, and market demand.

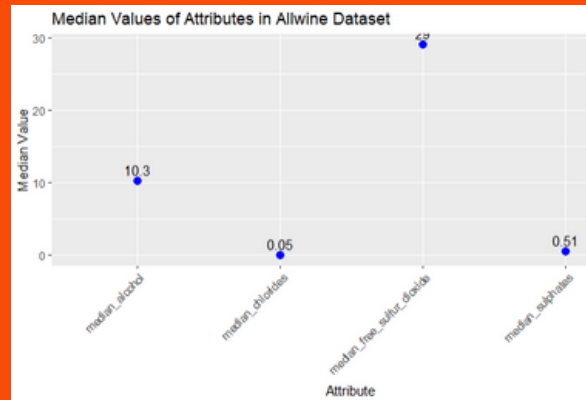
DESCRIPTIVE STATISTICS & VISUALIZATIONS ALL WINES

MEAN



This code reshapes the data resulting plot displays the mean values of each attribute as points with corresponding labels, aiding in the interpretation of central tendencies across different attributes.

MEDIAN



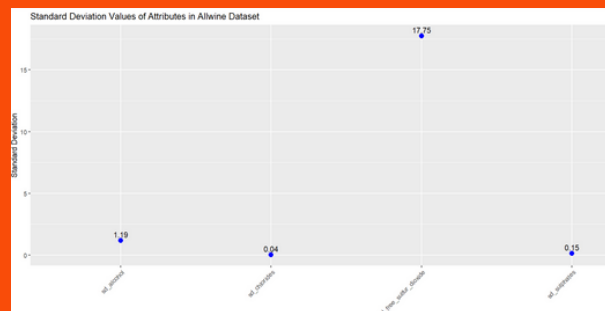
The median was calculated by determining the middle value in the dataset after arranging the values in ascending order

MODE



The points representing the mode values of different attributes in the "AllWine" dataset, with text labels indicating the mode values. The plot provides insights into the distribution of mode values across different attributes.

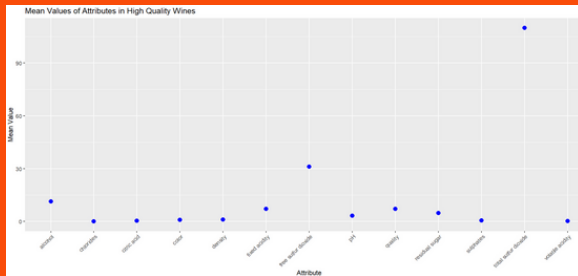
STANDARD DEVIATION



Each attribute is represented along the x-axis, while the corresponding standard deviation value is depicted along the y-axis.

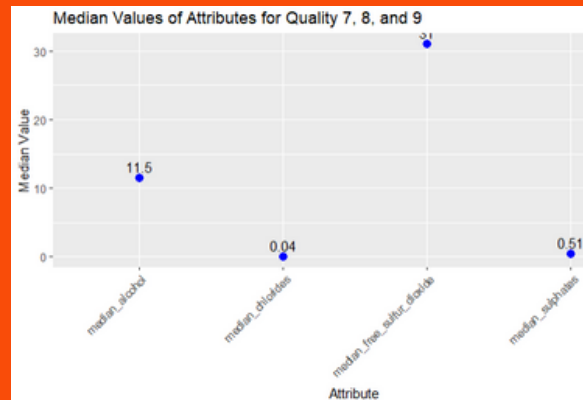
DESCRIPTIVE STATISTICS & VISUALIZATIONS WINE HIGH QUALITY

MEAN



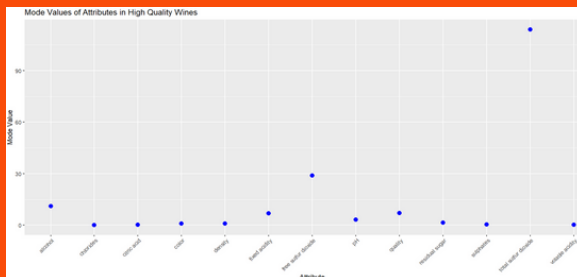
The mean was calculated by summing up all the values in the dataset and dividing by the total number of values.

MEDIAN



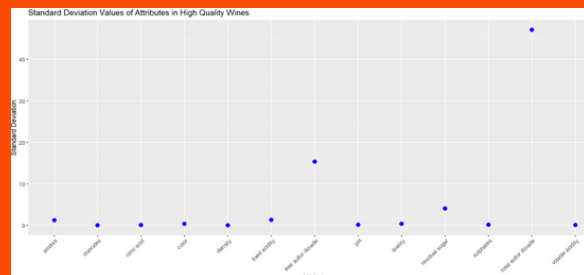
The median was calculated by determining the middle value in the dataset after arranging the values in ascending order.

MODE



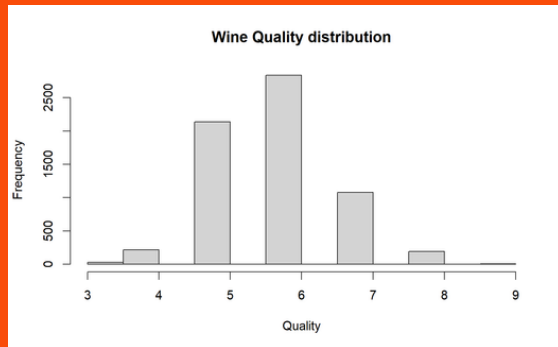
The mode was determined by finding the value that appeared most frequently in the dataset.

STANDARD DEVIATION

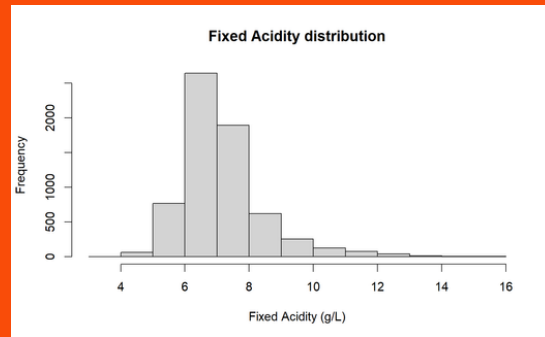


The standard deviation measures the amount of variation or dispersion of a set of values. It was calculated by taking the square root of the average of the squared differences between each value and the mean.

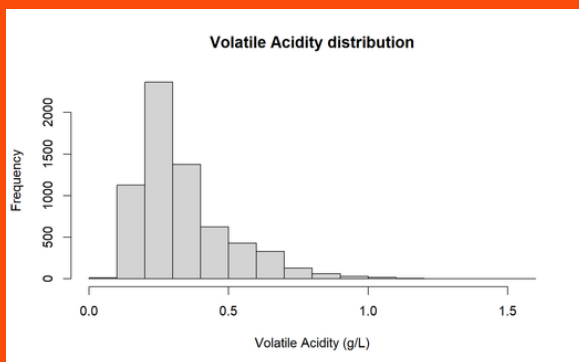
DESCRIPTIVE STATISTICS & VISUALIZATIONS ALL WINE



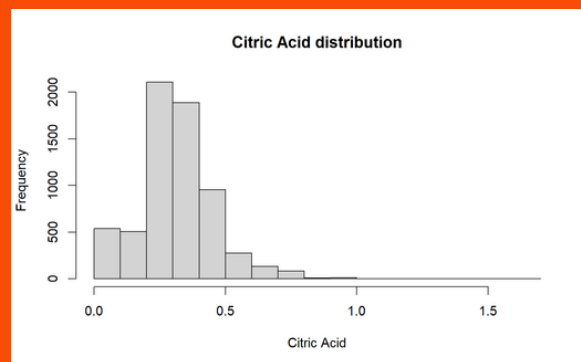
This Histogram shows the distribution of quality rating of all wines. A quality rating of a 6 is most frequent. This distribution appears to follow a normal distribution of quality ratings.



This Histogram shows the distribution of Fixed Acidity of all wines. This distribution appears to follow a normal distribution of quality ratings.

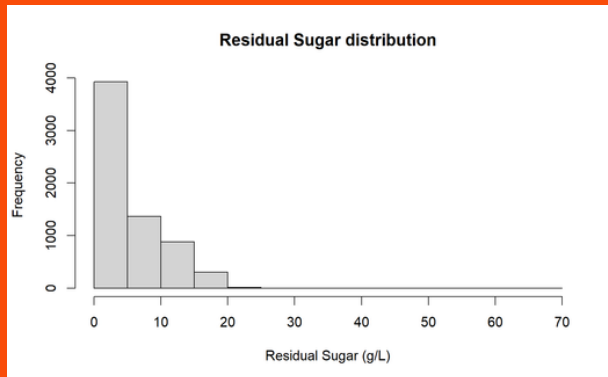


This Histogram shows the distribution of Volatile Acidity of all wines.

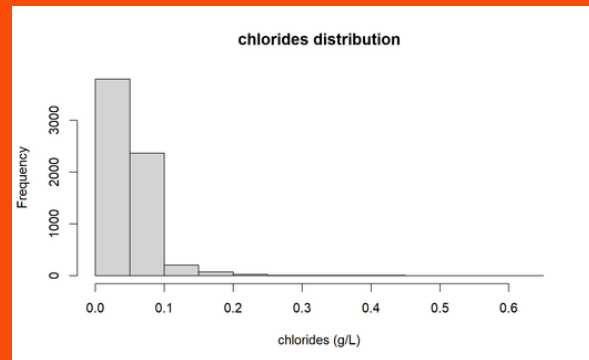


This Histogram shows the distribution of the amount of citric acid for all wines. A quality rating of a 6 is most frequent.

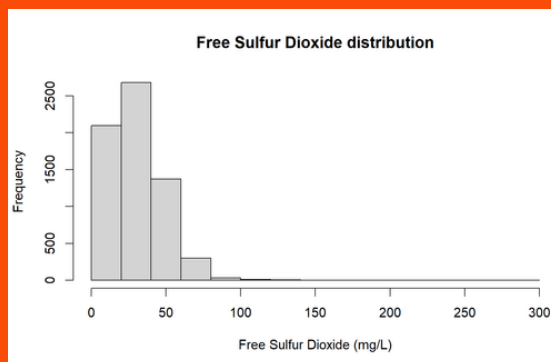
DESCRIPTIVE STATISTICS & VISUALIZATIONS ALL WINE



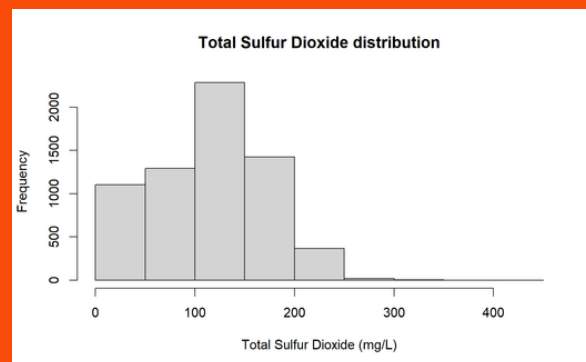
This Histogram shows the distribution of Residual Sugar for all wines. Clearly wines with less sugar are more common.



This Histogram shows the distribution of Chlorides for all wines.

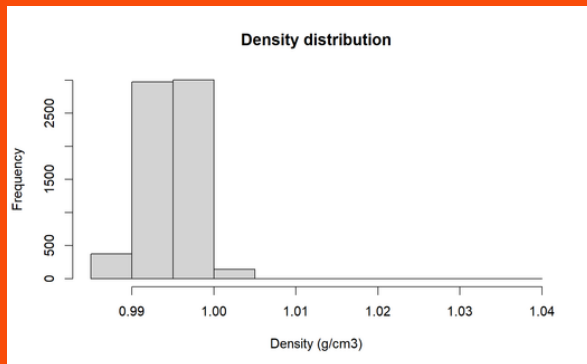


This Histogram shows the distribution of Free Sulfur Dioxide distribution for all wines.

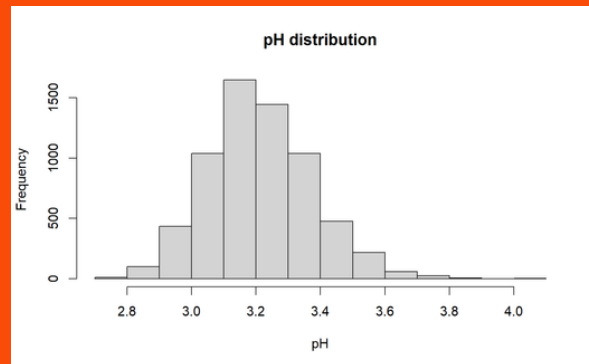


This Histogram shows the distribution of Total Sulfur Dioxide for all wines.

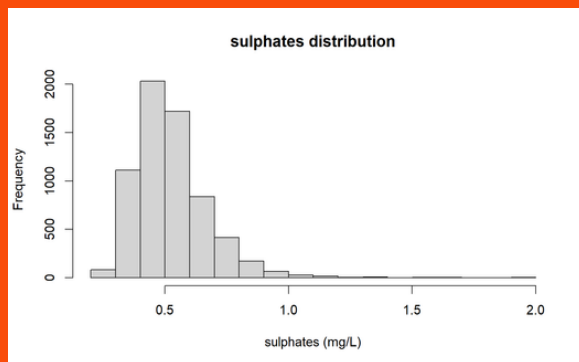
DESCRIPTIVE STATISTICS & VISUALIZATIONS ALL WINE



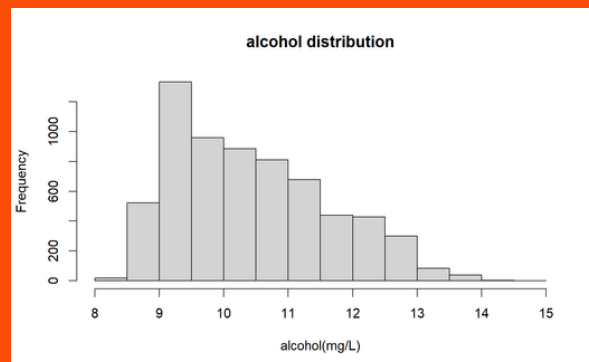
This Histogram shows the distribution of Density for all wines.



This Histogram shows the distribution of pH for all wines.

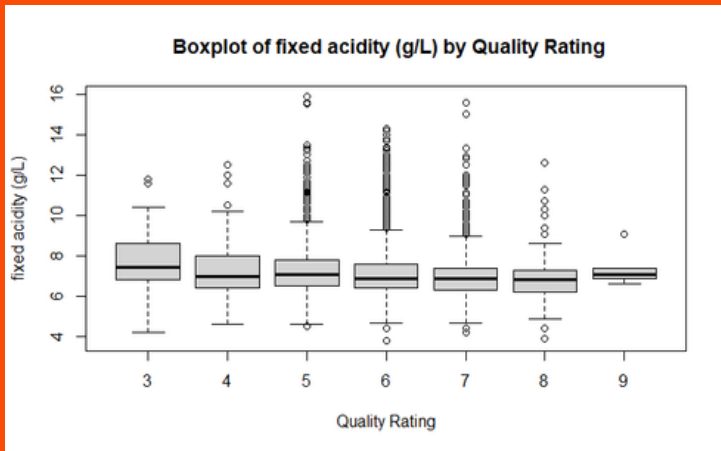


This Histogram shows the distribution of Free Sulfur Dioxide for all wines.

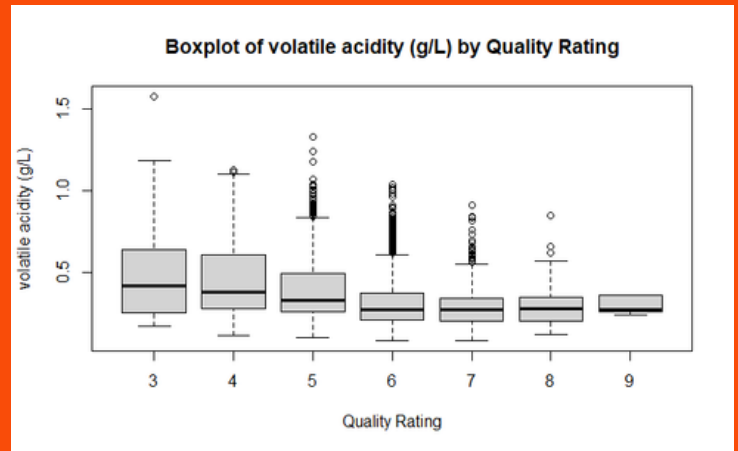


This Histogram shows the distribution of alcohol for all wines.

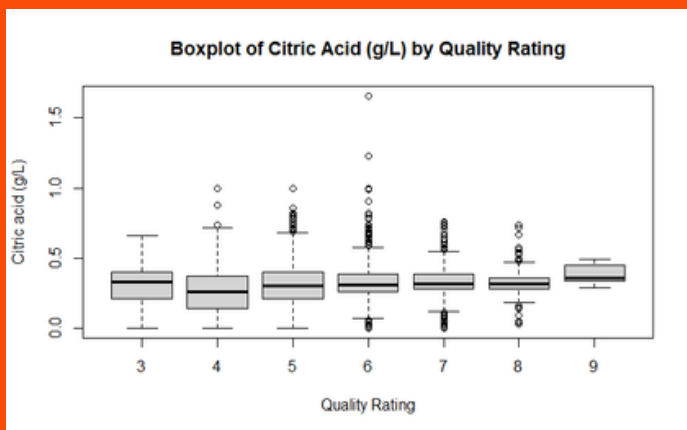
DESCRIPTIVE STATISTICS & VISUALIZATIONS ALL WINE



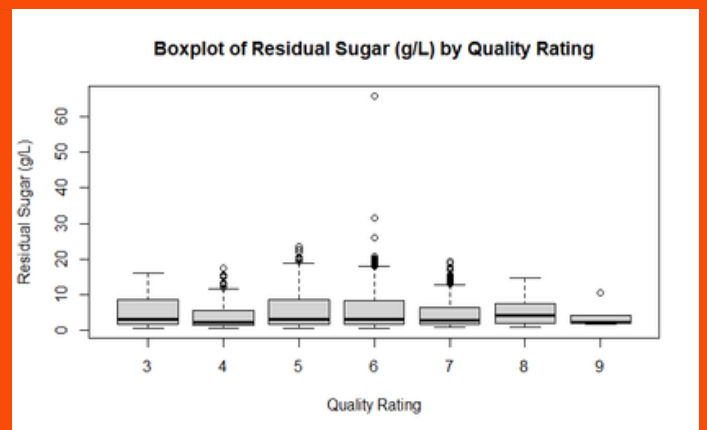
This Boxplot shows a lot of outliers at quality ratings of 5,6, and 7. A quality level of 9



This Boxplot shows a lot of outliers at quality ratings of 5,6, and 7. A quality level of 9

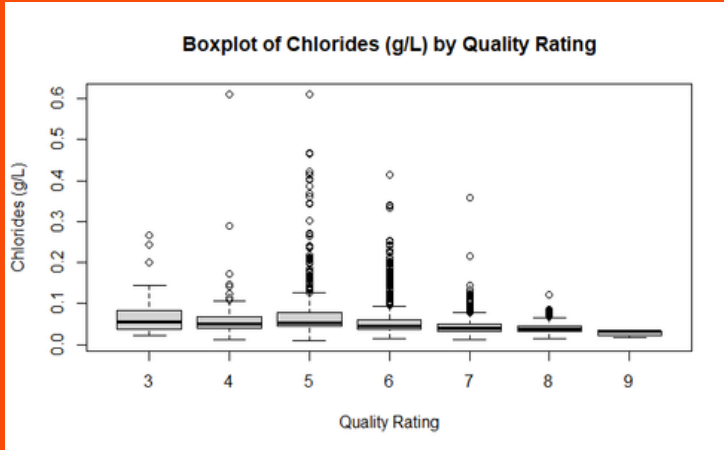


This Boxplot shows a lot of outliers at quality ratings of 5,6, 7, and 8. A quality level of 9

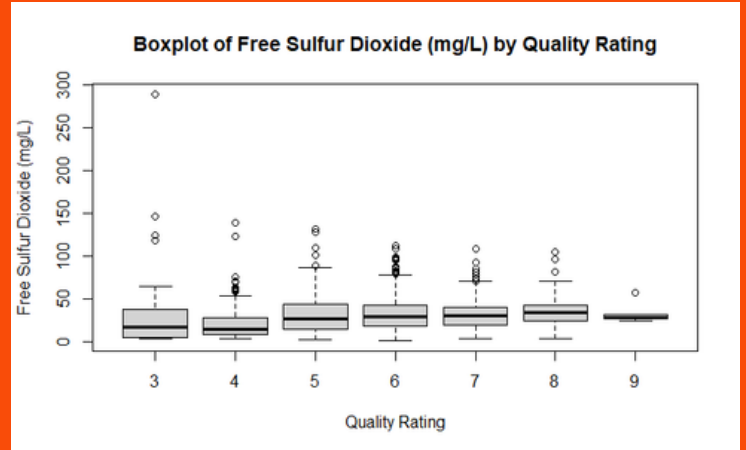


This Boxplot shows a lot of outliers at quality ratings of 4, 5,6, and 7. A quality level of 9

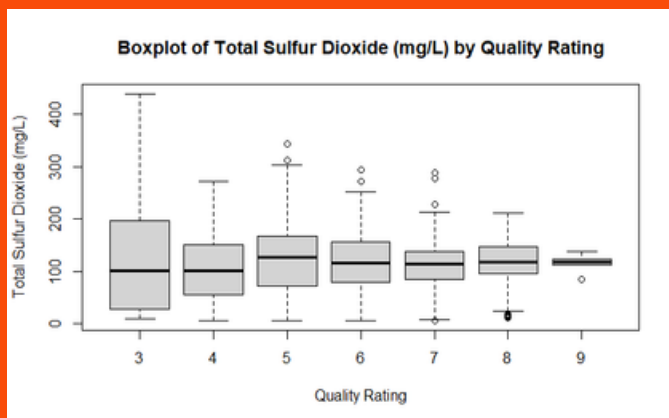
DESCRIPTIVE STATISTICS & VISUALIZATIONS ALL WINE



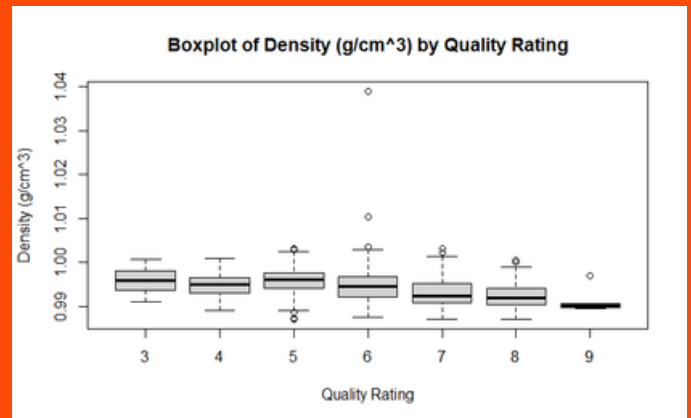
This Boxplot shows a lot of outliers at quality ratings of 5, 6, and 7. A quality level of 9



This boxplot shows moderate level of outliers across all of the Quality Ratings

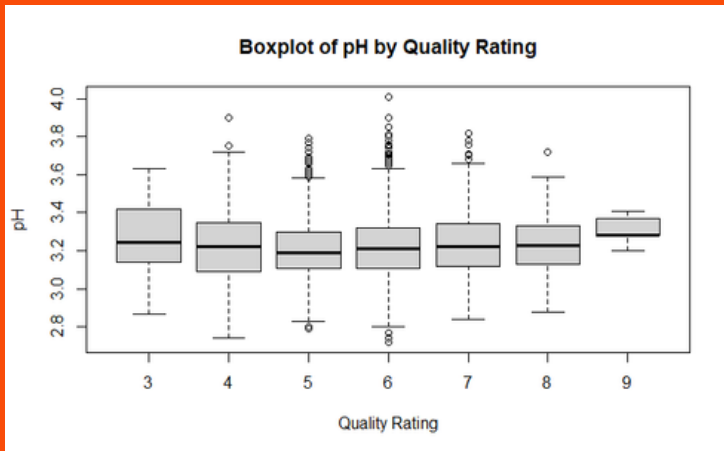


The range of Total Sulfide Dioxide Values decrease significantly as Quality Rating increases. This could be due to the due to more precise brewing practices associated with higher quality wine. One of the more consistent values for Quality Rating 6 and 7

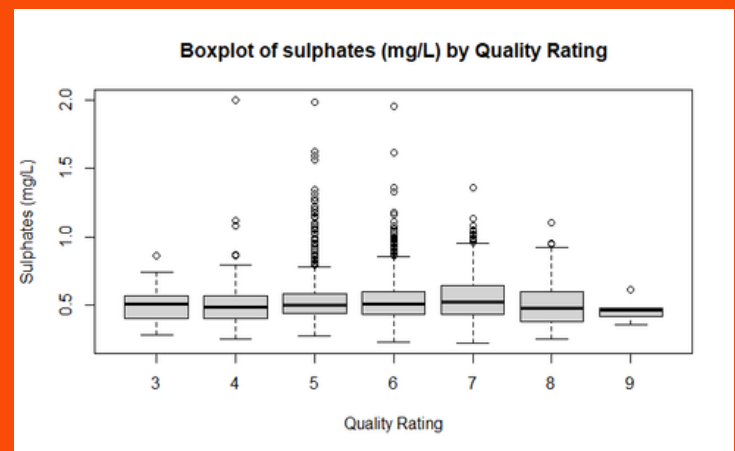


A quality rating of 9 has a very small range of values compared to the other ratings. You can see a slight decrease in values as the Quality Rating increases

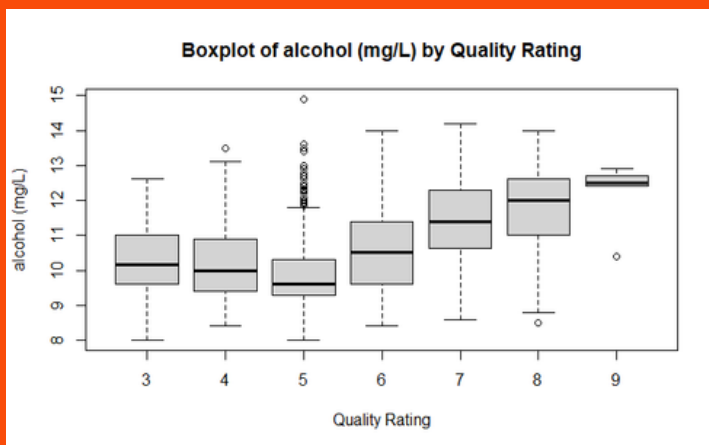
DESCRIPTIVE STATISTICS & VISUALIZATIONS ALL WINE



This Boxplot shows a lot of outliers at quality ratings of 5,6, and 7. A quality level of 9



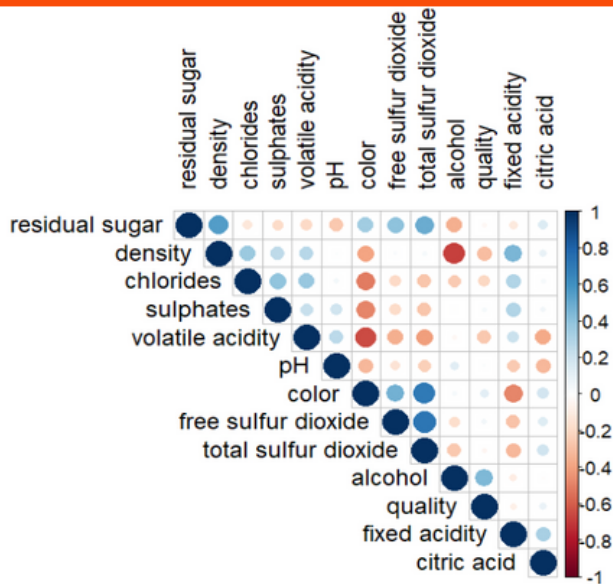
This Boxplot shows a lot of outliers at quality ratings of 5,6, and 7. A quality level of 9 has a very small range of values compared to the other ratings.



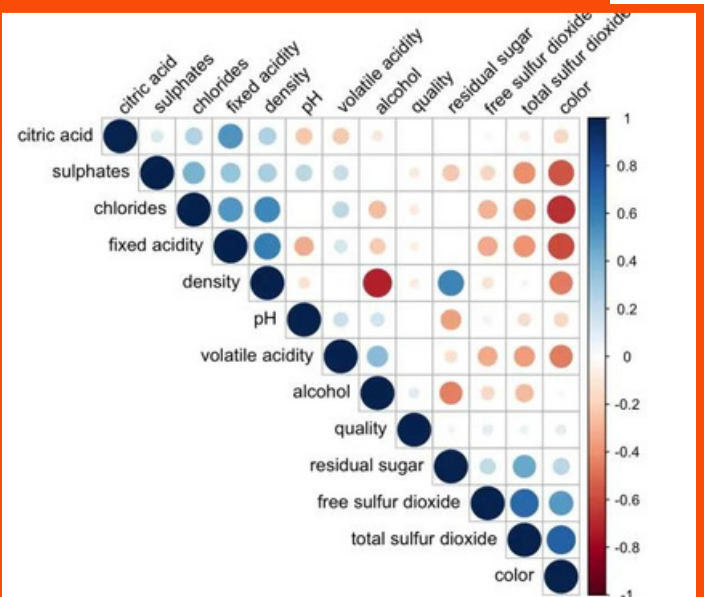
This Boxplot shows a lot of outliers at quality ratings of 5. There is clearly and upward trend in alcohol as the quality rating increases.

DATA MODELING & VISUALIZATIONS

ALL WINES

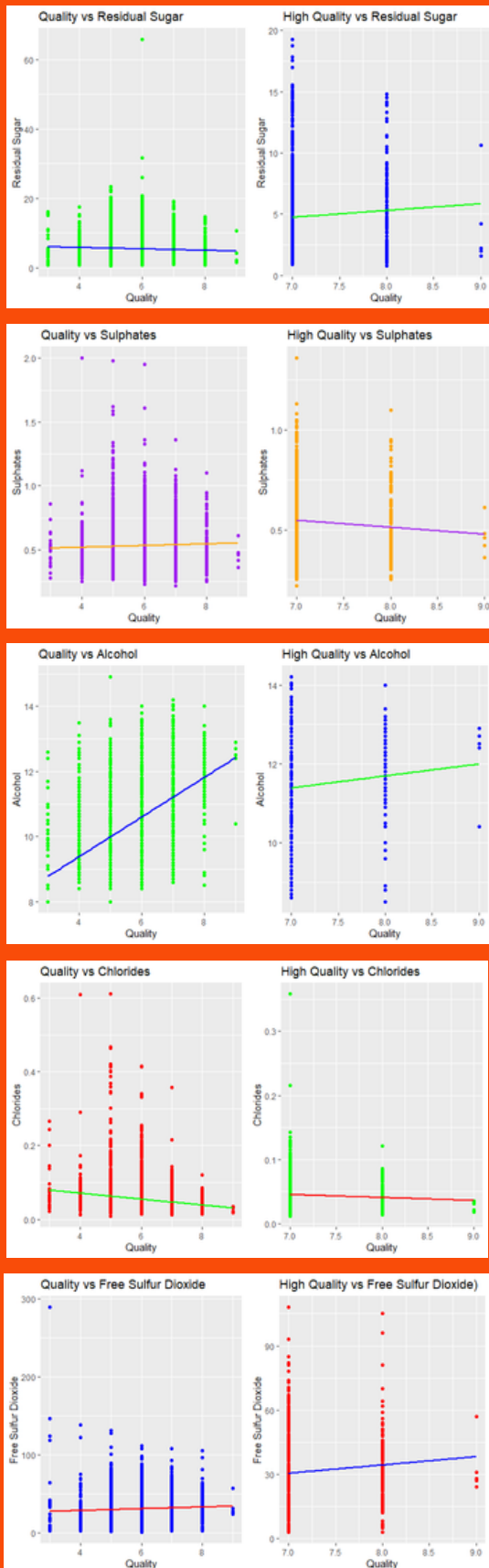


HIGH QUALITY



Each value in this vector indicates the degree of linear relationship between "quality" and a specific variable. When interpreting these correlation coefficients with values close to 1 or -1 indicate a strong correlation between the variables. A positive value suggests that as "quality" increases, the associated variable also tends to increase, while a negative value suggests an inverse relationship. Values close to 0 suggest a weak or negligible correlation between "quality" and the variable. These color and size gradients highlight strong correlations. For instance, we know citric acid has little to no significant relationship to the quality of wine as the correlogram shows a value of nearly 0.

DATA MODELING & VISUALIZATIONS

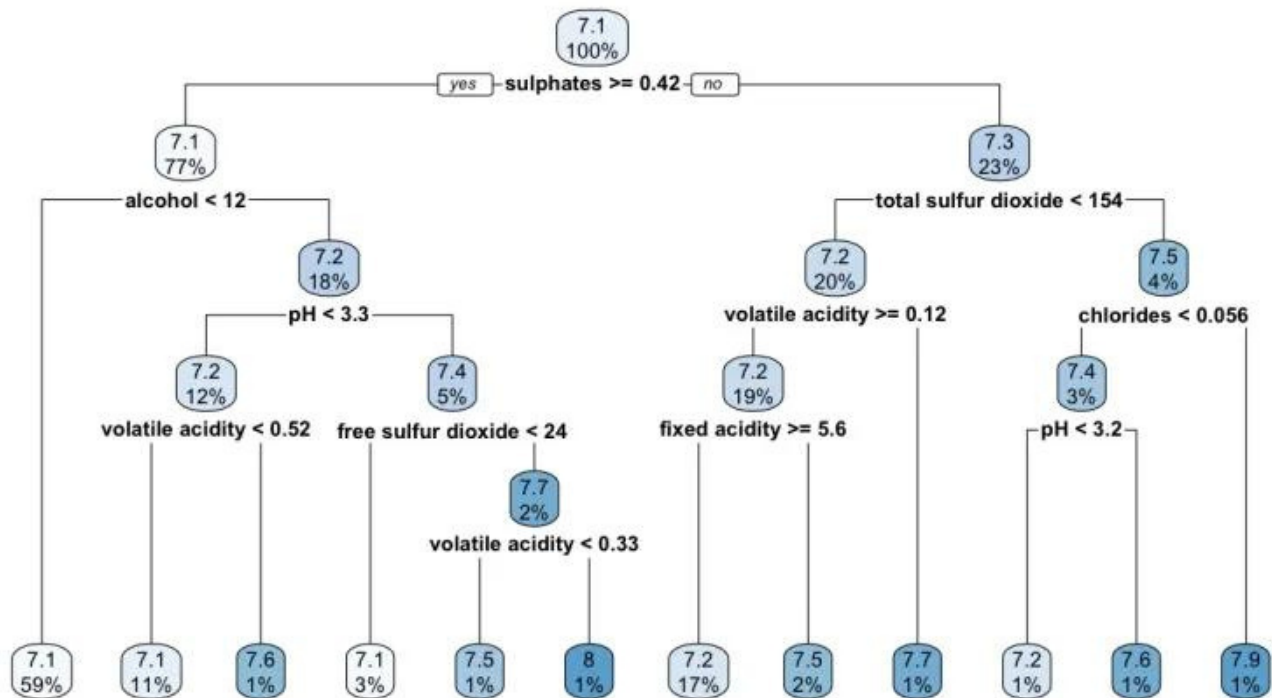


These plots delve into the relationships between Wine Quality and five different attributes. Residual Sugar, Sulphates, Alcohol levels, Chlorides, and Free Sulphur Dioxide. Each point is a Wine sample, with the linear regression line showing the trend of the models.

What is interesting to note is that, for the most part, all of the linear models tend to roughly follow the trend, with the exception of the Sulphates. the higher quality you go, the Linear Regression actually goes down.

DATA MODELING & VISUALIZATIONS

DECISION TREE



Tree model visualized from the Quality Wine Training Set. Looking at the sulphates level which determine if a wine is well preserved. We start here with 100% because all the wines in the set of quality wines don't have a lot of variance. We can see this as we follow the leaves down and see the wine ratings at the bottom.

Interpretation of The Results/Actionable Insights:

Q1: Differentiate the characteristics of high-quality red and white wines. What distinguishes a red wine as high-quality and a white wine as high-quality?

The results show these characteristics distinguish red and white wines as high quality:

Higher Alcohol Content: High-quality red wines tend to have higher alcohol content. For every one-unit increase in alcohol content, the quality of the wine is expected to increase by approximately 0.2227 units.

Higher Sulphates Content: Sulphates play a significant role in enhancing the quality of red wines. For each unit increase in sulphates, the quality of the wine is expected to increase by approximately 0.7217 units.

Lower Volatile Acidity: High-quality red wines typically have lower volatile acidity. For every one-unit increase in volatile acidity, the quality of the wine is expected to decrease by approximately 1.492 units.

Moderate to High Fixed Acidity: While fixed acidity positively correlates with wine quality, the relationship is relatively weaker compared to other factors. For each unit increase in fixed acidity, the quality of the wine is expected to increase by approximately 0.08507 units.

Moderate Residual Sugar: Residual sugar may vary but generally is moderate in high-quality red wines. For each unit increase in residual sugar, the quality of the wine is expected to increase by approximately 0.062 units.

Moderate Color Intensity: Red wines typically have a higher color intensity compared to white wines. However, excessive color intensity may negatively impact wine quality.

Interpretation of The Results/Actionable Insights: cont'd

Q2: What are the primary factors that influence the taste quality of red and white wines?

The results show the statistical significance of each attribute. Each attribute in bold shows that an increase in this attribute is an increase in wine quality:

Fixed Acidity: For each unit increase in fixed acidity, the quality of the wine is expected to increase by approximately 0.08507 units. This relationship is highly significant (p-value=7.05e-08).

Volatile Acidity: For each unit increase in volatile acidity, the quality of the wine is expected to decrease by approximately 1.492 units. This relationship is highly significant (p-value < 0.001).

Residual Sugar: An increase in residual sugar by one unit is associated with an increase in wine quality by approximately 0.062 units. This relationship is highly significant (p-value < 0.001).

Chlorides: Each unit increase in chlorides is associated with a decrease in wine quality by approximately 0.757 units. This relationship is statistically significant (p-value = 0.0236).

Free Sulfur Dioxide: An increase in each unit of free sulfur dioxide, wine quality is expected to increase by approximately 0.004937 units. This relationship is highly significant (p-value < 0.001).

Total Sulfur Dioxide: An increase in total sulfur dioxide by one unit is associated with a decrease in wine quality by approximately 0.001403 units. This relationship is highly significant (p-value < 0.001).

Density: Each unit increase in density is associated with a decrease in wine quality by approximately 103.9 units. This relationship is highly significant (p-value < 0.001).

pH: An increase in pH by one unit is associated with an increase in wine quality by approximately 0.4988 units. This relationship is highly significant (p-value < 0.001).

Sulphates: Each unit increase in sulphates is associated with an increase in wine quality by approximately 0.7217 units. This relationship is highly significant (p-value < 0.001).

Alcohol: An increase in alcohol content by one unit is associated with an increase in wine quality by approximately 0.2227 units. This relationship is highly significant (p-value < 0.001).

Color: Each unit increase in color intensity is associated with a decrease in wine quality by approximately 0.3613 units. This relationship is highly significant (p-value < 0.001).

Citric Acid: The coefficient for citric acid is -0.063, but it is not statistically significant (p-value = 0.4322). Therefore, there is no strong evidence to suggest a significant relationship between citric acid and wine quality in this analysis.

Interpretation of The Results/Actionable Insights: cont'd

Q2a: After identifying these factors, specify which one the company should prioritize to enhance or maintain the high-quality taste of each wine.

To prioritize which attribute a company should focus on to maintain high-quality taste of each wine, considering both the magnitude of the attribute and the significance level (p-value), the order from highest to lowest priority could be as follows:

1. Alcohol: It has both a highly significant coefficient and likely a substantial effect on wine quality, making it the highest priority for the company to optimize or maintain.
2. Sulphates: Similar to alcohol, sulphates exhibit a highly significant coefficient, suggesting a significant positive effect on wine quality. Thus, it should be a high priority for the company to manage sulphates levels effectively.
3. Residual Sugar: Despite having a positive effect on wine quality, residual sugar also demonstrates a highly significant coefficient. Thus, managing and potentially optimizing residual sugar levels could be the next priority.
4. Free Sulfur Dioxide and Total Sulfur Dioxide: Both variables have highly significant coefficients, indicating their substantial contribution to wine quality. Managing sulfur dioxide levels, both free and total, should be considered a significant priority.
5. Density: While density also shows a highly significant coefficient, its effect on wine quality is negative. Thus, minimizing density could be a priority, albeit less urgent compared to optimizing other factors.

CLEANING DATASETS TO REMOVE COMMAS, EXTRA SPACES, ETC.

```
# Load necessary libraries
library(readr)
# Load the dataset
allwines <- read.csv("C:/Users/leigh/Desktop/Syracuse/687/WineMerged.xlsx")
# Function to remove commas and spaces from all columns
clean_dataset <- function(data) {
  for (col in names(data)) {
    if (is.character(data[[col]])) {
      data[[col]] <- gsub(",", "", data[[col]]) # Remove commas
      data[[col]] <- gsub("\\s+", "", data[[col]]) # Remove spaces
    }
  }
  return(data)
}
# Cleanse the dataset
cleaned_allwines <- clean_dataset(allwines)
```

READING IN DATASETS INTO R

```
library(readxl)
#read in the datasets from the excel spreadsheets
red <- read_excel("/Users/zimrapanitz/Downloads/winequality-redfixed.xlsx")
white <- read_excel("/Users/zimrapanitz/Downloads/WineQuality-WhiteWine.xls")
```

ADDING A COLUMN TO IDENTIFY RED/WHITE WINE

```
#add column to each dataset to represent red or white wine
red$color <- c(0)
white$color <- c(1)
```

CREATING HISTOGRAMS OF RED/WHITE WINE BY QUALITY

```
#histogram of red wine quality
hist(red$quality, xlim = c(2, 10), breaks = 6, xlab="Quality", main="Histogram of Red Wine Quality")
#histogram of red wine quality
hist(red$quality, xlim = c(2, 10), breaks = 6, xlab="Quality", main="Histogram of Red Wine Quality")
```

MERGING TABLES TOGETHER INTO ONE DATASET "WINE"

```
library(tidyverse)
#add the tables together to create one larger table
wine <- rbind(red, white)
wine
head(wine)
tail(wine)
summary(wine)
color <- group_by(wine, color)
color
mean_quality <- summarise(color,type = mean(quality, na.rm = TRUE))
mean_quality
```


SPLITTING AND ARRANGING DATASET "WINE" BY QUALITY CATEGORIZED 7, 8, 9, 10 INTO "Quality_wine"

```
#Subset the dataset to include only rows where quality is 7, 8, or 9
Quality_wine <- allwine[allwine$quality %in% c(7, 8, 9),]
#Select only numeric columns
qualitynum <- Quality_wine[sapply(Quality_wine, is.numeric)]
#Calculate correlations
correlation_matrix <- cor(qualitynum)
#Sort correlations with quality
cor_with_quality <- correlation_matrix["quality",]
cor_with_quality <- cor_with_quality[order(abs(cor_with_quality), decreasing = TRUE)]
#Display correlations
print(cor_with_quality)
```

CALCULATE MEDIAN VALUES FOR EACH ATTRIBUTE FOR QUALITY LEVELS 7, 8, 9

```
median_values <- Quality_wine %>%
  filter(quality %in% c(7, 8, 9)) %>%
  summarise(
    median_free_sulfur_dioxide = median(free sulfur dioxide),
    median_alcohol = median(alcohol),
    median_chlorides = median(chlorides),
    median_sulphates = median(sulphates))
```

PLOTTING MEDIAN VALUES OF ATTRIBUTES IN ALLWINE DATASET USING POINT GEOM

```
#Select only numeric columns in the allwine dataset
allnumeric <- allwine[sapply(allwine, is.numeric)]
#Calculate median values for each attribute for quality levels 7, 8, and 9
median_values <- allnumeric %>%
  summarise(
    median_free_sulfur_dioxide = median(free sulfur dioxide),
    median_alcohol = median(alcohol),
    median_chlorides = median(chlorides),
    median_sulphates = median(sulphates) )
#Reshape data for plotting
median_values_long <- tidyr::pivot_longer(median_values,
  cols = everything(),
  names_to = "Attribute",
  values_to = "Median")
```

PLOTTING MEDIAN VALUES OF ATTRIBUTES FOR ALLWINES USING POINT GEOM

```
ggplot(median_values_long, aes(x = Attribute, y = Median)) +
  geom_point(stat = "identity", color = "blue", size = 3) +
  geom_text(aes(label = round(Median, 2)), vjust = -0.5) +
  labs(x = "Attribute", y = "Median Value", title = "Median Values of Attributes in Allwine Dataset") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ylim(0, NA) # To remove the lower bound of y-axis
```

PLOTTING MEDIAN VALUES OF ATTRIBUTES FOR QUALITY 7, 8, 9, 10 USING POINT GEOM

```
#Plot
ggplot(median_values_long, aes(x = Attribute, y = Median)) +
  geom_point(stat = "identity", color = "blue", size = 3) +
  geom_text(aes(label = round(Median, 2)), vjust = -0.5) +
  labs(x = "Attribute", y = "Median Value", title = "Median Values of Attributes for Quality 7, 8, and 9") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ylim(0, NA) # To remove the lower bound of y-axis
```

PLOTTING MODE VALUES OF ATTRIBUTES FOR ALLWINES USING POINT GEOM

```
ggplot(mode_values_long, aes(x = Attribute, y = Mode)) +
  geom_point(stat = "identity", color = "red", size = 3) +
  geom_text(aes(label = Mode), vjust = -0.5) +
  labs(x = "Attribute", y = "Mode Value", title = "Mode Values of Attributes in Allwine Dataset") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ylim(0, NA) # To remove the lower bound of y-axis
```

PLOTTING MODE VALUES OF ATTRIBUTES FOR QUALITY 7, 8, 9, 10 USING POINT GEOM

```
ggplot(mode_df, aes(x = Attribute, y = Mode)) +
  geom_point(color = "blue", size = 3) +
  labs(x = "Attribute", y = "Mode Value", title = "Mode Values of Attributes in High Quality Wines") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

PLOTTING SD VALUES OF ATTRIBUTES FOR ALLWINES USING POINT GEOM

```
ggplot(standard_deviation_values_long, aes(x = Attribute, y = Standard_Deviation)) +
  geom_point(stat = "identity", color = "blue", size = 3) +
  geom_text(aes(label = round(Standard_Deviation, 2)), vjust = -0.5) +
  labs(x = "Attribute", y = "Standard Deviation", title = "Standard Deviation Values of Attributes in Allwine Dataset") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

PLOTTING SD VALUES OF ATTRIBUTES FOR QUALITY 7, 8, 9, 10 USING POINT GEOM

```
ggplot(sd_df, aes(x = Attribute, y = Standard_Deviation)) +
  geom_point(color = "blue", size = 3) +
  labs(x = "Attribute", y = "Standard Deviation", title = "Standard Deviation Values of Attributes in High Quality Wines") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
str(HighQuality_Wines)
```

PLOTTING MEAN VALUES OF ATTRIBUTES FOR ALL WINES USING POINT GEOM

```
ggplot(mean_values_long, aes(x = Attribute, y = Mean)) +
  geom_point(stat = "identity", color = "blue", size = 3) +
  geom_text(aes(label = round(Mean, 2)), vjust = -0.5) +
  labs(x = "Attribute", y = "Mean Value", title = "Mean Values of Attributes in Allwine Dataset") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

PLOTTING MEAN VALUES OF ATTRIBUTES FOR QUALITY 7, 8, 9, 10 USING POINT GEOM

```
ggplot(mean_df, aes(x = Attribute, y = Mean)) +
  geom_point(color = "blue", size = 3) +
  labs(x = "Attribute", y = "Mean Value", title = "Mean Values of Attributes in High Quality Wines") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

PLOTTING VALUES USING GEOM_BOXPLOT**PLOTTING CORRELOGRAM MODEL USED FOR ALL WINES AND QUALITY WINES >= 7**

```
corrplot(Allcorrelation, method = "circle", type = "upper", order = "hclust",
  tl.col = "black", tl.srt = 45,)
correlation_matrix <- cor(qualitynum)
cor_with_quality <- correlation_matrix["quality", ]
cor_with_quality <- cor_with_quality[order(abs(cor_with_quality), decreasing = TRUE)]
```

PLOTTING DECISION TREE MODEL FOR QUALITY 7, 8, 9, 10

```
####Tree model for quality with the Training set
table(TestingW$quality)
tree_model <- rpart(quality ~ ., data = TrainingW, method = "anova")
tree_model
rpart.plot(tree_model)
```

PLOTTING LINEAR MODEL FOR ALLWINES

```
ggplot(AllWines, aes(x = X1, y = X2)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(x = "alcohol", y = "chlorides", title = "Linear Visualization for AllWines Dataset") +
  THEME_MINIMAL()
```

PLOT FOR QUALITY VS FREE SULFUR DIOXIDE WITH LINEAR REGRESSION LINE

```
plot_1a <- ggplot(allwine, aes(x = quality, y = free sulfur dioxide)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") + # Add linear regression line
  labs(x = "Quality", y = "Free Sulfur Dioxide", title = "Quality vs Free Sulfur Dioxide (with Linear Regression)")
```

PLOT FOR QUALITY VS ALCOHOL WITH LINEAR REGRESSION LINE

```
plot_2a <- ggplot(allwine, aes(x = quality, y = alcohol)) +
  geom_point(color = "green") +
  geom_smooth(method = "lm", se = FALSE, color = "blue") + # Add linear regression line
  labs(x = "Quality", y = "Alcohol", title = "Quality vs Alcohol (with Linear Regression)")
Plot for quality vs chlorides with linear regression line
plot_3a <- ggplot(allwine, aes(x = quality, y = chlorides)) +
  geom_point(color = "red") +
  geom_smooth(method = "lm", se = FALSE, color = "green") + # Add linear regression line
  labs(x = "Quality", y = "Chlorides", title = "Quality vs Chlorides (with Linear Regression)")
```

PLOT FOR QUALITY VS SULPHATES WITH LINEAR REGRESSION LINE

```
plot_4a <- ggplot(allwine, aes(x = quality, y = sulphates)) +
  geom_point(color = "purple") +
  geom_smooth(method = "lm", se = FALSE, color = "orange") + # Add linear regression line
  labs(x = "Quality", y = "Sulphates", title = "Quality vs Sulphates (with Linear Regression)")
```

PLOT FOR QUALITY VS FREE SULFUR DIOXIDE WITH LINEAR REGRESSION LINE

```
plot_1b <- ggplot(qualitynum, aes(x = quality, y = free sulfur dioxide)) +
  geom_point(color = "red") +
  geom_smooth(method = "lm", se = FALSE, color = "blue") + # Add linear regression line
  labs(x = "Quality", y = "Free Sulfur Dioxide", title = "High Quality vs Free Sulfur Dioxide (with Linear Regression)")
```

PLOT FOR QUALITY VS ALCOHOL WITH LINEAR REGRESSION LINE

```
plot_2b <- ggplot(qualitynum, aes(x = quality, y = alcohol)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", se = FALSE, color = "green") + # Add linear regression line
  labs(x = "Quality", y = "Alcohol", title = "High Quality vs Alcohol (with Linear Regression)")
```

PLOT FOR QUALITY VS CHLORIDES WITH LINEAR REGRESSION LINE

```
plot_3b <- ggplot(qualitynum, aes(x = quality, y = chlorides)) +
  geom_point(color = "green") +
  geom_smooth(method = "lm", se = FALSE, color = "red") + # Add linear regression line
  labs(x = "Quality", y = "Chlorides", title = "High Quality vs Chlorides (with Linear Regression)")
```

PLOT FOR QUALITY VS SULPHATES WITH LINEAR REGRESSION LINE

```
plot_4b <- ggplot(qualitynum, aes(x = quality, y = sulphates)) +
  geom_point(color = "orange") +
  geom_smooth(method = "lm", se = FALSE, color = "purple") + # Add linear regression line
  labs(x = "Quality", y = "Sulphates", title = "High Quality vs Sulphates (with Linear Regression)")
grid.arrange(plot_1a, plot_1b, ncol = 2)
grid.arrange(plot_2a, plot_2b, ncol = 2)
grid.arrange(plot_3a, plot_3b, ncol = 2)
grid.arrange(plot_4a, plot_4b, ncol = 2)
```

THANK YOU